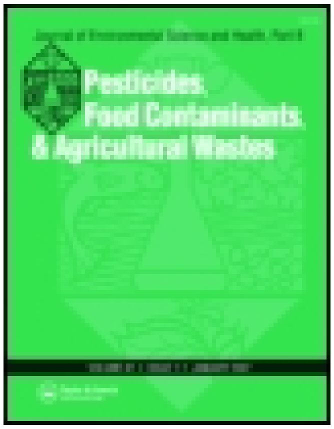

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The effects of soil and *Trifolium repens* (white clover) on the fate of estrogen

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In this study, we investigated the behavior of estrogens in the rhizosphere of white clover (*Trifolium repens*, clover hereafter) with two different pot tests, using soil and agar as growth media. In a pot test using agar spiked with estrogen, the estrogen concentration in the agar with clover decreased to non-detectable levels within one month, while in the agar without clover, 60% of initially added estrogen remained after one month. The half-lives of estrone (E1) and 17β-estradiol (E2) in the agar with clover were 2.4–3.8 and 13.2 d, respectively. The dissipation of E1 followed first-order rate law, while that of E2 fitted a zero-order reaction, indicating that they had different mechanisms of dissipation from agar. In the soil pot test, the behavior of E1 and E2 was not influenced by clover. An initial rapid decrease in the amount of estrogen extracted by methanol/acetic acid was followed by persistence for 1–3 months, regardless of presence of clover. Moreover, in three weeks E1 and E2 were only partly degraded by microbes extracted from the soil used in the pot test. In this study, abiotic degradation of estrogens and sorption of estrogen to soil, rather than the effects of soil microbes and clover, contributed to the initial rapid dissipation of estrogens in the soil. However, the results of the agar pot test suggested that vegetation such as clover may significantly contribute to removal of estrogens when estrogens in aqueous phase are discharged with surface runoff and preferential flow after heavy rain in agricultural fields, or when present in soils with low estrogen sorptivity.

Keywords: Estrogen; estrone; 17β-estradiol; white clover; soil pot test; agar pot test; estrogen dissipation; livestock manure; dissipation kinetics.

Introduction

There is increasing concern that the steroid estrogens, including estrone (E1), 17β-estradiol (E2), and estriol (E3), derived from livestock manure can act as endocrine disruptors and adversely affect aquatic ecosystems. Among the steroidal hormones, E1 and E2 in particular may have a significant impact on the environment, as both compounds are abundant in livestock manure. Reportedly, almost all estrogens excreted in pregnant cow feces are E1 and E2 in unconjugated form.[1]

Estrogens have a relatively short half-life in agricultural soils.[2–5] In a study using ^{14}C-labelled steroid compounds in manure applied to soil, Lucas and Jones[5] concluded that risk of estrogen contamination to aquatic life was very low at normal landspreading rates because the biodegradation half-life of the hormones was 5–25 d.

There are many reports on surface runoff and leaching from manure-treated agricultural fields contributing estrogens to surface water and groundwater.[6–9] Kjær et al.[9] found that E1 and E2 were leached from manure-treated structured soils to drainage water for three months in concentrations exceeding the lowest observable effect levels of 3.3 and 14 ng/L, respectively. Lai et al.[10] implied that estrogens discharged into rivers could be degraded rapidly because they remain in the aqueous rather than the solid phase due to competition with other estrogens and other hydrophobic chemicals.

The research cited above established very well the behavior of estrogens as a consequence of their application to agricultural soils; however, there are few reports on effects of plants on the fate of estrogens. The oxidation of aqueous estrogens due to enzymes such as peroxidase exuded from horseradish (*Armoracia rusticana*) and

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laccase exuded from mushroom (*Agricus bisporus*) has been reported.[11–13] The enhanced degradation of estrogens in the presence of pasture and agricultural crops can be inferred from these observations. Among agricultural plants, white clover (*Trifolium repens*, clover hereafter) is a primary pasture plant for livestock. As a legume, clover can fix nitrogen and provides benefits for soils through fertilization as well as mineralization of xenobiotics by rhizodeposit enhancement.[14] Clover can be a very useful plant in remediation of endocrine-disrupting substances including estrogens in the rhizosphere.

The behavior of estrogens in soils can be governed by complex processes such as soil sorption, microbial immobilization and mineralization peculiar to soil microbes. Consequently, the sterilized agar was used as a control growth medium in this study to clarify the effect of clover on the fate of estrogen in the absence of soil.

The purpose of this study was to investigate the effects of plants on the fate and dissipation rate of E1 and E2 in the rhizosphere using soil and agar planted with clover.

Materials and methods

Reagents and analytical procedures

All chemicals were of Japan Industrial Standard special grade or better, purchased from Wako Pure Chemical Industries Ltd (Osaka, Japan) unless stated otherwise. *N*, *O*-Bis(trimethylsilyl)trifluoroacetamide + 1% Trimethylchlorosilane (BSTFA + 1% TMCS) was purchased from Pierce (Rockford, Ill., USA).

Solid phase extraction (SPE) on C18 silica was conducted to separate a group of organic components including estrogens from inorganic components and solvents. 17β-estradiol (16, 16, 17-d_3), hereafter d-E2, was used as a surrogate compound. After washing 200–2000 mg of C18 resin packed in a glass column with a few column volumes of dichloromethane, methanol and Milli-Q water, 50–80 mL of the aqueous sample spiked with 10 or 100 μL of 10 mg/L d-E2 methanol solution was loaded, followed by a hexane wash to remove the residual water. Then the SPE cartridge was dried under a gentle stream of nitrogen, and estrogens eluted with 6–10 mL of methanol/ethyl acetate (1:5, v/v) to obtain 6 mL of elutant. If necessary a clean-up using aminopropyl resin was conducted to remove the color of the elutant. The elutant was evaporated to dryness at 40°C in a vial (Agilent Technologies, Germany) under nitrogen gas and stored at −6°C until derivatization. Estrogen derivazation was with 100 μL of BSTFA + 1% TMCS for 90 min at 90°C. The derivatized samples were evaporated to dryness and redissolved in acetone before a gas chromatograph mass spectrometer (GC/MS) analysis. The recovery of d-E2 in SPE was 73 ± 3% (average ± standard error, n = 60).

Analytical instruments

A Shimadzu GC/MS-QP5050 equipped with a Restek Rtx-5MS capillary column (Bellefonte, PA, USA), 30 m × 0.25 mm × 0.25 μm film thickness, was employed to analyze estrogens. The carrier gas was helium at a flow rate of 30 mL/min, using splitless injection. The GC oven temperature program was started at 50°C and held for 1 min, then ramped to 230°C at 20°C/min (no hold), then to 320°C at 5°C/min and held for 2 min. The total run time was 32 min, and the injector was operated at 280°C. A mass spectrometer equipped with an electron impact source (energy approximately 70 eV) was used as the detector. E1 and E2 were monitored at mass number 343 and 416, respectively, in the selected ion monitoring (SIM) mode, and a four-point calibration curve (either 0, 0.1, 0.5 and 1 mg/L or 0, 1, 5 and 10 mg/L) used to quantify E1, E2 and d-E2. Chrysene-d_{12} was used as an internal standard. The injection volumes of standards and samples were 1 μL.

Pot test using soil spiked with estrogen

The soil was amended with either E1 or E2 alone at a concentration of 10 or 100 μg/kg dry-soil. Clover was grown for three months on the E1-spiked soil pot test and one month on the E2-spiked soil pot. A pot with estrogen-spiked soil without clover was prepared as control.

The soil used in this study was an Andosols (according to FAO/UNESCO classification) from Shobara, Hiroshima Prefecture, Japan, which was granulated using bentonite and water. The soil had been used as a sorbent in a rapid infiltration system to treat wastewater from a livestock farm prior to this study.[15] Soil properties are shown in Tables 1 and 2. Nutrients measured were phosphate, extractable calcium (ECa), extractable magnesium (EMg) and extractable potassium (EK). Phosphate (Trg-P) was determined by Truog's method.[16] ECa, EMg and EK were determined from 1 M ammonium acetate extraction of soil. The extract was analyzed using an atomic absorption spectrometer (Z-6100, Hitachi Ltd, Tokyo, Japan) for EK, and an inductively coupled plasma (ICP) atomic emission spectrometer (ICPS-1000TR, Shimadzu Co, Kyoto, Japan) for ECa and EMg. Total carbon (TC) and total nitrogen (TN) of the soil were determined by dry combustion method (PE 2400 series CHNS/O analyzer, PerkinElmer Japan, Kanagawa, Japan).

The soil was pulverized, gravel removed and air-dried prior to use. To prepare the E1- or E2-spiked soil, 0.1 or 1 mL of 100 mg/L-methanol E1 or E2 solution and 4 mL of water were added per 1 kg dry-soil, thoroughly mixed with a spade, and dried overnight in the laboratory to make the estrogen concentrations 10 or 100 μg/kg dry-soil.

Pots of different sizes were used to accommodate plants after varying growing periods. In the E2-spiked soil test, 0.3-L (12.5 cm diameter × 8.2 cm height) pots were used for the one-month test. In the E1-spiked soil test, 0.7-L

Table 1. Physical and chemical properties of soil used in this study.

				Nutrient level			
Bulk density (g/cm^3)	*Specific surface area (m^2/g)*	*Soil pH (1:5 H_2O)*	*Soil EC (1:5 H_2O) (mS/cm)*	*Trg-P (P_2O_5 mg/kg dry-soil)*	*ECa (CaO mg/kg dry-soil)*	*EMg (MgO mg/kg dry-soil)*	*EK (K_2O mg/kg dry-soil)*
1.21	5.4	7.0	0.065	183	4170	782	930

(12.4 cm diameter × 13.1 cm height), 1.3-L (15.2 cm diameter × 16.2 cm height), 7-L (25 cm × 25 cm × 23.8 cm) and 13-L (30 cm × 30 cm × 28.8 cm) pots were used according to experiment duration. Of each pot, 70–80% of volume was filled with estrogen-spiked or control soil. Approximately 30 sprouts of clover (seeds purchased from Takii Seed Co Ltd, Kyoto, Japan.), grown on the Andosols (not spiked with estrogens) for 10–14 d, were transplanted into each pot. The E1-spiked pots were placed in a 25°C homeothermic room (SPH-24S2, Sakai Engineering Co Ltd, Japan) for 2, 4, 8, 10 and 12 weeks. The E2-spiked pots were placed in the same room for 7, 26 and 33 d. The clover was watered almost every day with 50–200 mL of Milli-Q water according to pot size. Light was fluorescent with a color spectrum similar to sunlight (Nisso Co Ltd, Japan). After each period, the pots were dismantled and the rhizosphere soil collected.

Extraction of estrogens from soil

Extraction of water-soluble estrogens from soil

Samples of 20 g of moist soil stored at 4°C were extracted with 80 mL of Milli-Q water by shaking for 24 h on a horizontal shaker (120 rpm) in a 100-mL glass screw vial. Extracts were filtered with GF/F glass fiber filters (Whatman International Ltd, Tokyo, Japan) and analyzed for estrogens after C18 SPE.

Extraction of solvent-soluble estrogens

The extraction procedure was basically the same as used for analysis of estrogens in sediments by the Japan Sewage Works Association.[17] After water-soluble estrogen extractions, soils were air-dried at ambient temperature in a desiccator. Then 7.5 g of dried soil was placed in a 50-mL Teflon® centrifuge tube (Nalgene) and extracted with 30 mL of methanol/1M acetic acid (9:1, v/v) by shaking the tube for 30 min on the shaker. The supernatant was obtained by centrifugation at 3000 rpm and 4°C for 15 min. The residues were further shaken with 30 mL of 100% methanol for 5 min and the supernatant was collected as previously. The two extracts were mixed, filtered with a GF/F filter, and were concentrated on a rotary evaporator (Tokyorikakikai Co Ltd, Japan) at 40°C to less than 10 mL. The concentrate was diluted to 200 mL with Milli-Q water and analyzed for estrogen after C18 and aminopropyl SPE.

Table 2. Total nitrogen (TN) and TOC contents of soil used in this study.

TN (%)	*TOC (%)**	*C/N ratio*
0.58	9.80	18.05

*Total organic compound (TOC) was estimated from the result of TC measurement, assuming that there was little calcium carbonate in the sample.

Investigation on the effect of soil microbes on estrogen dissipation

To investigate the effect of microbes from the soil used in this study on estrogen dissipation, a biodegradation test was conducted by adding E1 or E2 at 100 μg/L in soil solution and measuring estrogen concentrations 1, 3, 7, 14, 17 and 21 d after the start of the experiment.

To obtain soil microbes, 10 g of moist soil was shaken with 100 mL of Milli-Q water for 1.5 h on a horizontal shaker (160 rpm) in a 100-mL glass screw vial, and stood overnight in a homeothermic room at 25°C. The supernatant was filtered with 2.0 μm pore diameter Isopore Millipore membrane filters. Milli-Q water spiked with each estrogen was prepared for controls. Adenosine triphosphate (ATP) concentrations (measured using ATP-analyzer AF-100, DKK-TOA Co, Japan) were 300 pmol/L in soil solution in 9 after extraction of soil microbes and below detection level (2 pmol/L) in Milli-Q water stored in the homeothermic room at 25°C for 22 d.

Pot test using agar spiked with estrogen

A Hoagland's solution containing potassium nitrate (5 mmol/L), potassium phosphate monobasic (1 mmol/L), calcium nitrate (5 mmol/L) and magnesium sulfate (2 mmol/L) was spiked with E1 or E2 (methanol solution) to final concentrations of 20 or 200 μg/L. The spiked Hoagland's solution was solidified by adding an equal volume of 0.8% (w/w) agar solution, which was previously sterilized at 121°C for 30 min, and cooled to 70°C. Agar-solidified Hoagland's solution (hereafter agar) without estrogens was prepared as control. In the final agar medium, the methanol concentration was adjusted to 1%.

The clover seeds (0.2 g) were spread on 500 mL of agar medium in 500-mL glass beakers and grown without watering for two or four weeks. Photoperiods and temperatures were light/dark 10/14 h at 10–15°C and 14/10 h at 20–25°C for the four-week test, and 14/10 h at 20–25°C for the two-week test. Clover height was measured to indicate plant growth. After the test, agar samples were centrifuged for 24 h at 2900 rpm and 4°C, the supernatants collected and filtered with Durapore or Omnipore Millipore membrane filters of 0.2 μm pore diameter, and analyzed by GC/MS.

Dissipation kinetics of estrogens in the agar

The dissipation kinetics and the half-life of each estrogen were determined from the temporal variation of estrogen concentrations in the agar medium.

Results and discussion

Estrogen concentrations in the E1-spiked soil pot test

No water-soluble E1 fractions were detected in any soil samples collected after different incubation periods. Furthermore, there was no indication of E2 production from E1 by reductive reaction in the water-soluble estrogen fractions.

The solvent-soluble E1 concentrations decreased to <20% of the initial concentrations within two weeks of incubation (Table 3), and became constant 56–84 d after the incubation commenced, regardless of presence of clover. The concentrations of solvent-soluble E1 were 5% of initial concentration after 84 d. Similar rapid decreases in E1 concentration have been reported many times.[2,5,18] As shown in Table 2, the soil used in the pot test was high in organic matter (ca. 10% w/w organic carbon). Generally, soil organic matter has a high affinity for nonpolar organic compounds and dominates their sorption to soil when there is >3% organic matter in soil.[19] It is therefore probable that E1 was irreversibly sorbed to the Andosols by physicochemical mechanisms.

In short, there was no effect of clover on the concentration of water-soluble and solvent-soluble E1 in the soil. Irreversible physicochemical sorption of estrogen on soil particles and/or the microbial degradation rather than phytoextraction and/or phytodegradation of estrogen by clover had more influence on the decrease in solvent-soluble E1 in the soil. In other words, the plant availability of E1 was low in this soil.

Table 3. Solvent-soluble E1 concentrations in the soil (μg/kg dry-soil) after different periods of incubation in the E1-spiked soil.

	Time after incubation (d)					
	Initial	*14*	*28*	*56*	*70*	*84*
Clover	10	1.4	1.2	1.0	ND	1.2
Clover	100	9.8	22.9	6.9	6.0	5.1
Control (without clover)	100	18.8	16.4	8.0	4.2	4.2

ND: not detected.

Estrogen concentrations in the E2-spiked soil pot test

Water-soluble estrogens were not detected in the E2-spiked soil, except for the samples collected 19 d after starting incubation (0.85–1.50 μg/L in the interstitial water). The estrogen concentrations in the interstitial water were estimated from the water-soluble estrogen concentrations and water content of soil samples.

In the E2-spiked soil pot test there was a rapid decline within 7 d, followed by a gradual decrease in the solvent-soluble E2 concentrations (Table 4). E1 as an oxidative product of E2 was detected within 7 d when clover was present, E1 concentrations increased with time along with oxidation of E2, probably due to soil microbes. There ware similar increases in E1 concentrations in biodegradation tests of E2 (Fig. 1). Consequently, the concentration of solvent-soluble E1 exceeded that of E2 at the end of the experiment. The differences in the total estrogen concentrations were not statistically significant between soil with and without clover. Therefore, in the E2-spiked soil pot test, the dissipation of estrogens due to plants was low, suggesting that clover had little effect on the fate of either E1 or E2 in this soil.

There is much literature on estrogen extraction from soil.[4,20–24] Chun et al.[23] reported that all E2 could not be extracted from soil because of a number of intra-soil processes. Van Emmerik et al.[22] showed that most E2 sorbed on goethite, kaolinite, and illite could be easily extracted by water or methanol; but E2 sorbed in montmorillonite was not, suggesting that E2 was incorporated into interlayer spaces. These studies indicate that the estrogen extractability depended on the type of soil minerals, and that estrogen may not be quantitatively extracted from some soils. Therefore, in the present study, we used the same extraction procedure on every soil sample and observed the variation in the percentages of estrogen extracted rather than trying to extract all estrogen from the soil.

Estrogen concentrations in the biodegradation test

Estrogen concentrations in the soil solution spiked with E1 or E2 are shown in Figure 1. There was oxidation of E2 to E1, reduction of E1 to E2, and degradation of E1 after 7 d. The results suggest a lag phase in the reactions of estrogens mediated by soil microbes.

In the E2-spiked biodegradation test, 40% of E2 was oxidized to E1 by soil microbes 7 d after the start of the test, but total estrogens (E1 + E2) remained almost constant (Fig. 1). Total estrogen concentration was 80% of the initial concentration after 21 d.

E1 and E2 were not completely degraded by the soil microbes within a few weeks. Contrary to our results,

Table 4. Solvent-soluble E2 and E1 concentrations in the soil (μg/kg dry-soil) during different periods of incubation in the E2-spiked soil pot test.

	Time after incubation (d)									
	0	*7*			*26*			*33*		
	E2	*E1*	*E2*	*E1 + E2*	*E1*	*E2*	*E1 + E2*	*E1*	*E2*	*E1+E2*
White clover	10	ND	1.0	1.0	0.3	0.5	0.8	0.5	0.4	0.9
White clover	100	3.2	21.2	24.3	2.2	8.4	10.6	6.1	4.4	10.5
Control (without clover)	100	ND	17.1	17.1	2.5	7.4	9.9	5.7	5.5	11.2

ND: Not detected.

transformation of E2 to E1 was rapid and the half-life of E2 almost identical to that of E1 (1–9 d) in manure-amended soils.[5] This indicates that the rate of oxidation of E2 to E1 may differ depending on species of soil microbes.

The soil pot and biodegradation tests enable the inference that the rapid decrease of E1 and E2 in estrogen-spiked soil tests was not caused by soil microbes.

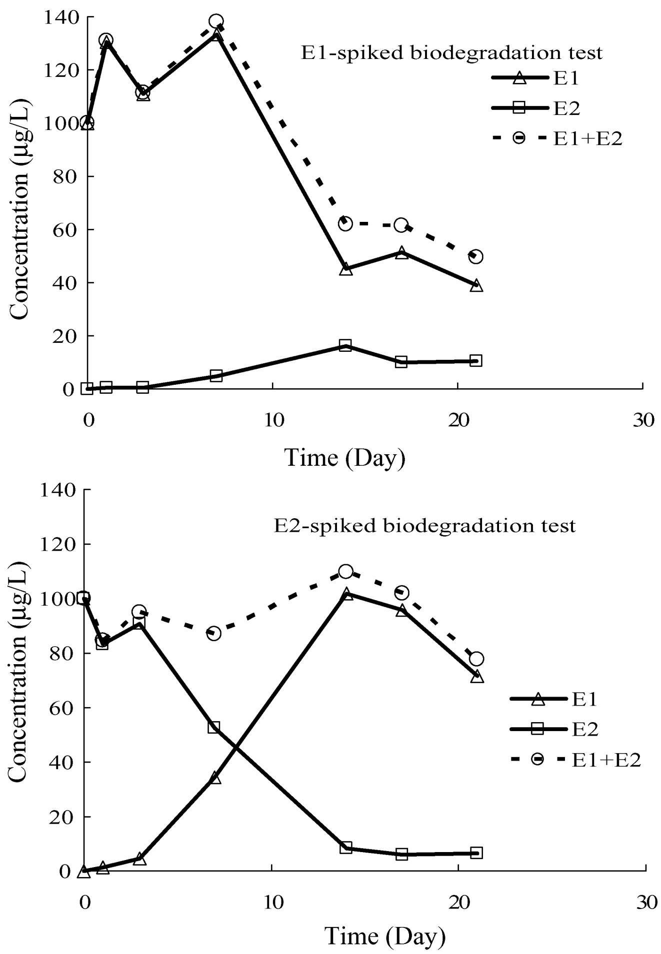

Fig. 1. Estrogen concentrations in the aqueous phase containing soil solution spiked with each estrogen.

Estrogen concentrations in the E1 or E2 spiked agar tests

The estrogen concentrations in agar at the end of the experiment (four weeks) at air temperatures of 10–15°C and 20–25°C are given in Tables 5 and 6, respectively. E1 + E2 concentrations were > 60% of the initial concentrations (10 or 100 μg/L) in agar without clover.

In the agar with clover, E1 and E2 concentrations were significantly decreased to several percent of initial concentrations, regardless of initial estrogen form. The estrogen concentrations were lower in the agar with clover at air temperature of 20–25°C compared to 10–15°C. In contrast, in agar without clover, there was no difference in the estrogen concentrations at different temperatures. Clover grew much more slowly at 10–15°C than at 20–25°C (Fig. 2). These results suggest that dissipation of estrogens in the agar depended on clover growth.

Lee and Liu[25] reported that sewage bacteria could degrade five metabolites of E2 (E1, E3, 16α-hydroxyestrone, 2-methoxyestrone and 2-methoxyestradiol) via ring cleavage. Soil microbes may degrade steroid estrogens similarly to sewage bacteria, but they were absent in the agar. The decreased estrogen concentrations in agar with clover are attributable to estrogen uptake by clover,[26] plant metabolism,[27] sorption onto roots, or degradation by certain enzymes exuded by roots[28] (hereafter; plant effects).

Horseradish peroxidase (HRP) and phenoloxidase are used to remove estrogens from municipal wastewater in laboratory-scale experiments.[11–13] Monophenol monooxygenase (EC 1. 14. 18. 1) and phenol 2-monooxygenase (EC 1. 14. 13. 7) are found in tissue extracts and root exudates of alfalfa (*Medicago sativa* L.), white mustard (*Sinapis alba* L.) and cress (*Lepidium sativum* L.).[29] To our knowledge, however, there is no information on any clover enzymes that degrade estrogens. Further research concerning the dissipation pathway of estrogen in the presence of clover is needed. Both in controls and agar with clover (Tables 5 and 6), there was reduction of E1 to E2 and oxidation of E2 to E1. Although oxidation of E2 to E1 is well-known, the reduction of E1 to E2 is less common and may be explained by fungi growing on agar, which can exude 17β-hydroxysteroid dehydrogenases

Table 5. Concentrations of estrogens (μg/L) at the end (four weeks) of the estrogen-spiked agar test at air temperature 10–15°C.

	E1		*E2*	
Initial	*Clover*	*Control (without clover)*	*Clover*	*Control (without clover)*
E1 10 μg/L	0.57 ± 0.49	7.39	0.10 ± 0.14	2.72
E2 10 μg/L	0.30 ± 0.02	1.98	0.60 ± 0.01	5.16
E1 100 μg/L	0.55 ± 0.50	73.58	< 0.04	15.90
E2 100 μg/L	1.13 ± 0.70	14.61	0.18 ± 0.04	72.84

*Values are corrected for reduction in the agar volume due to evaporation and transpiration. Values represent means ± SD (n = 2), except control (n = 1).

known for reversible interconversions of 17-hydroxy and 17-keto steroids.[30]

Dissipation kinetics of E1 and E2 in agar

The E1 or E2 concentrations in agar normalized to initial concentrations (Fig. 3) were almost identical irrespective of initial estrogen concentrations (10 or 100 μg/L) when the initial form of estrogen was the same.

There was a significant difference in the relative estrogen concentrations in agar depending on the type of estrogens added. E2 was removed more slowly than E1 when clover was present. However, both E1 and E2 were almost completely removed from agar after 28 d. In the E1-spiked agar, the change in the relative concentration of E1 (R[E1]) as a function of time (t) fitted a first-order reaction with $R^2 > 0.97$. The formula for E1 dissipation was described by $R[\mathrm{E1}] = \exp(-0.2114t)$ and $R[\mathrm{E1}] = \exp(-0.2882t)$ for 10 and 100 μg/L of initial E1, respectively. The half-life of added E1 was 2.4–3.8 d. This is consistent with reports by Auriol et al.,[12,13] who reported that HRP-catalyzed reaction and laccase-catalyzed reaction of E1 exhibited pseudo-first-order dependence on estrogen concentrations.

In the E2-spiked agar, apparent dissipation and oxidation of E2 resembled a zero-order rather than a first-order reaction; a Michaelis–Menten mechanism and/or heterogeneous enzymatic catalysis are suggested as a cause.

When substrate concentration is much higher than Michaelis constant, the kinetics of the Michaelis–Menten mechanism on enzymic reactions approximates a zero-order reaction, which is described by the following formula: $|d[\mathrm{E2}]/dt| = k_b[\mathrm{Enz}]$, where [E2] and [Enz] represent the E2 concentration and the total concentration of enzyme, respectively, and k_b is the reaction rate constant. The half-life $t_{1/2}$ of the zero-order reaction is $t_{1/2} = [\mathrm{E2}]_0/k_b[\mathrm{Enz}]$. Here $t_{1/2}$ is proportional to the initial E2 concentration $[\mathrm{E2}]_0$ in the E2-spiked agar test. However, the half-life of E2 added to the agar was 13.2 d regardless of the initial E2 concentration (10 or 100 μg/L). This result indicated that the enzyme participating in E2 dissipation was exuded in proportion to the initial E2 concentrations.

On the other hand, heterogeneous enzymatic catalysis is based on the Langmuir approach of an ideal surface, which stems from a mechanism allowing one relatively rapid (e.g. adsorption equilibrium) reaction followed by a single, slow surface-reaction step. Langmuir kinetics can also approximate a zero-order reaction due to complete covering of active sites by the substrate (e.g. E2) in cases of high substrate concentration.[31] Zero-order kinetics of E2 dissipation may occur when the enzyme of E2 dissipation is distributed only on the surface of a root completely covered with excessive E2. Further work is required to identify the detailed mechanism of E2 dissipation in the presence of clover.

Comparisons between soil–plant and agar pot tests

The effect of clover on the behavior of estrogen in the rhizosphere differed significantly between the soil pot and the agar pot tests. In agar, E1 and E2 concentrations were significantly decreased when clover was present, while there was little degradation of E1 and E2 in agar without clover.

Table 6. Concentrations of estrogens (μg/L) at the end (four weeks) of the estrogen-spiked agar test at air temperature 20–25°C.

	E1		*E2*	
Initial	*Clover*	*Control (without clover)*	*Clover*	*Control (without clover)*
E1 10 μg/L	< 0.03	5.17 ± 0.67	< 0.02	1.31 ± 0.47
E2 10 μg/L	< 0.03	2.42 ± 2.66	< 0.02	1.19 ± 0.82
E1 100 μg/L	0.01 ± 0.02	80.58 ± 3.23	< 0.01	12.42 ± 0.29
E2 100 μg/L	0.02 ± 0.03	40.51 ± 13.44	0.01 ± 0.02	18.87 ± 6.53

*Values are corrected for reduction in the agar volume due to evaporation and transpiration. Values represent means ± SD (n = 2). The detection levels are subject to sample volume used.

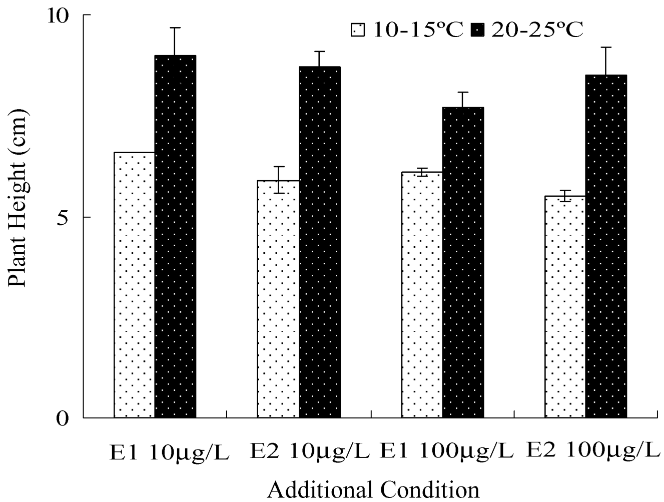

Fig. 2. Effect of different air temperatures on clover height grown on agar medium.

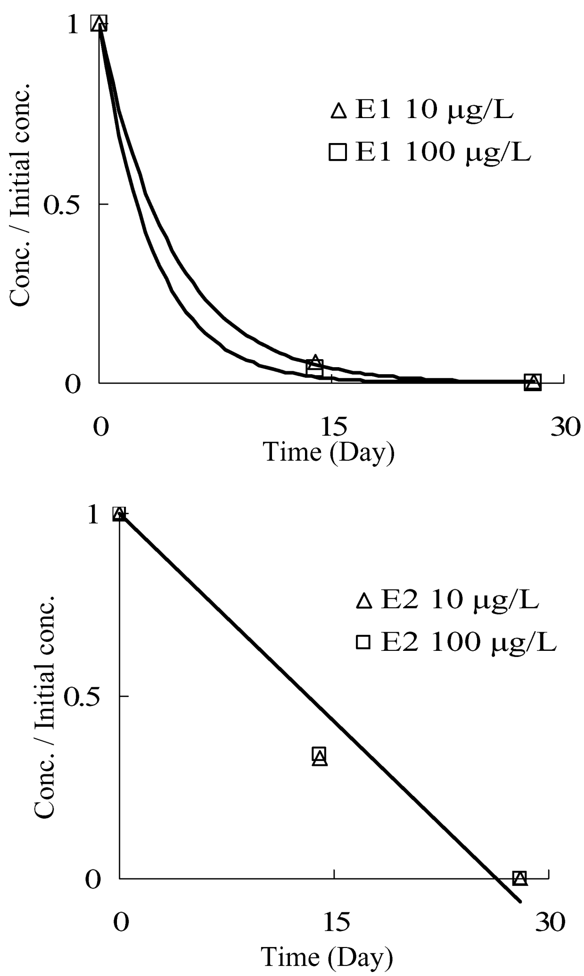

Fig. 3. Degradation of each estrogen compound in the agar medium in the presence of clover. Initial estrogen concentrations are 10 (triangle) or 100 (square) μg/L (n = 2).

The results showed that estrogens in the plant rhizosphere were dramatically decreased due to plant effects.

The half-lives of E1 and E2 in agar were estimated as 2.4–3.8 and 13.2 d, respectively, showing that E2 was more persistent than E1 in agar with clover. As mentioned previously, estrogens (E1 + E2) in the E2-spiked soil solution degraded more slowly than those in the E1-spiked soil solution, due to a lag phase in the oxidation of E2 to E1. In short, E2 in the liquid phase was less degradable than E1 in the presence of soil microbes or clover.

In the estrogen-spiked soil pot test, the presence of clover did not enhance dissipation of estrogens, and there was no lag phase in the oxidation of E2 (Tables 3 and 4).

It is also probable that any enzymes from the plant root was sorbed to the soil, and inactivated. For example, Claus and Filip[32] reported that tyrosinase, laccase and peroxidase were strongly sorbed to bentonites and clay–humus complex and that most of the enzymes lost their activity when adsorbed in bentonite. On the other hand, Serban and Nissenbaum[33] found that humic acid association with enzymes such as peroxidase and catalase formed an activated complex. Thus we speculated that enzyme activity in the soil can depend heavily on the kind of enzymes and the type and components of soils used, or that soil sorption of estrogens could have restricted their transport to the plant root, thereby limiting the effects of phytodegradation and/or phytoextraction processes on the fate of estrogens.

The soil pot test indicated that the plant availability of estrogens in this soil was extremely low.

Conclusion

We investigated the rate of E1 and E2 dissipation in soil and agar medium with and without clover. The concentrations of both E1 and E2 were drastically decreased in agar within one month when clover was present, indicative of plant effects. The degree of dissipation of estrogens in agar was dependent on plant growth. In contrast, estrogen behavior in the Andosols was more affected by estrogen sorption on soil and abiotic degradation of estrogens than by clover.

The dissipation rate and reaction type in agar differed between E1 and E2. The dissipation rate of E1 fitted a first-order-reaction, as in the case of HRP- and laccase-catalyzed degradation of E1. However, the dissipation kinetics of E2 was zero-order, and the half-life of E2 was independent of the initial estrogen concentration. Data suggested that a Michaelis–Menten mechanism was involved, where any enzymes involved in E2 dissipation were exuded proportionally to the initial E2 concentration and/or that there was heterogeneous enzymatic catalysis on the root surface. Further research is necessary to elucidate in detail the mechanism that retards the dissipation of E2.

Our study revealed that although clover can remove the steroid hormones from the aqueous media, it could not remove estrogens from the Andosols. However, estrogen effluence from agricultural land reportedly occurs as preferential flow in soil and surface runoff after storm events. Also, in soils with low sorptivity of estrogens or soils with few soil microbes, estrogens may persist in the interstitial water. Under such circumstances, estrogen removal from the aqueous phase using vegetation such as clover may be an effective method to prevent the pollution of aquatic environments. Complete removal of estrogens from the aqueous phase using clover took approximately one month.

A detailed and comprehensive pathway of estrogen degradation in the presence of clover is required to evaluate its usefulness and to build a strategy for use in estrogen removal.

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